# Notes: Jiang, Peng, and Yan (JFE, 2024)

## Personality Differences and Investment Decision-Making

## Contents

# 1 Big picture

- **Question.** Can persistent personality traits explain heterogeneity in investor beliefs, risk preferences, social interaction tendencies, and equity allocations?
- **Main object.** The paper studies the Big Five personality traits: Agreeableness, Conscientiousness, Neuroticism, Extraversion, and Openness.
- **Primary data.** The authors design and administer a nationwide survey through the American Association of Individual Investors (AAII), yielding 3,325 valid responses from affluent, experienced U.S. investors.
- **Core conceptual framework.** Portfolio choice is partly standard mean-variance optimization and partly a pull toward a non-pecuniary "target portfolio." Personality can affect both channels.
- **Belief result.** Personality traits explain meaningful cross-sectional variation in stock return expectations, crash beliefs, GDP growth expectations, and inflation expectations. Neuroticism is the standout belief-related trait.
- **Risk preference result.** Openness, Extraversion, and Agreeableness are strongly related to risk aversion. Higher Openness and higher Extraversion correspond to lower risk aversion; higher Agreeableness corresponds to higher risk aversion.
- **Belief formation result.** Neurotic investors are more mean-reverting and less extrapolative in stock trend beliefs, while open investors are more extrapolative and less mean-reverting.
- **Social interaction result.** Neuroticism and Extraversion predict stronger willingness to follow popular investments among people around the respondent.
- **Portfolio result.** High Neuroticism and low Openness are associated with lower equity allocations in the AAII sample.
- **Mechanism result.** Neuroticism appears to matter primarily through pessimistic beliefs and crash worries; Openness appears to matter primarily through risk tolerance and the target portfolio/nonstandard channel.
- **Out-of-sample evidence.** Results replicate in representative household panels from Australia (HILDA) and Germany (GSOEP). Neuroticism is negatively related to equity investment and Openness is positively related.
- **Takeaway.** Personality traits are not just demographic controls. They are persistent noncognitive attributes that organize beliefs, preferences, social behavior, and portfolio choice.

**Read the paper.**

- The paper is best read as a bridge between psychology and household finance: personality traits are used to organize persistent heterogeneity in beliefs, preferences, social motives, and portfolio outcomes.
- The main empirical claim is not that personality replaces standard economics. Rather, personality helps explain where subjective beliefs, risk aversion, and non-pecuniary investment motives come from.
- The key distinction is between *mechanisms* and *reduced-form portfolio effects*. Neuroticism matters because it predicts pessimism and crash worries; Openness matters because it predicts risk tolerance and broader willingness to engage with financial-market uncertainty.
- The strongest result is the recurring two-trait pattern: Neuroticism predicts lower equity exposure and Openness predicts higher equity exposure in the AAII survey and in representative household panels from Australia and Germany.
- The evidence is organized sequentially: first measure personality; then link personality to beliefs, risk aversion, and social interaction; then ask whether these variables help explain the final portfolio allocation.

# 2 Data and measurement

## 2.1 Survey sample: AAII investors

- The authors administer the main survey through the American Association of Individual Investors (AAII), a nonprofit organization with roughly 150,000 members.
- The survey was emailed on November 22, 2019, with a reminder on November 29.
- The final sample contains 3,325 valid responses after filtering.
- The response rate is about 2%, but the respondent pool is especially relevant for investment decisions because AAII members are experienced individual investors.
- Median reported wealth is $3.5 million; about 90% have a college degree; more than 80% have wealth greater than $1 million.

**Interpretation:** The AAII sample is not representative of the full U.S. population. It is intentionally focused on affluent market participants. This makes the results especially useful for studying active household financial decision-making among investors with meaningful portfolios.

## 2.2 Big Five personality traits

The survey uses a 20-item questionnaire, with four items for each Big Five trait. Responses are used to construct five personality scores.

- **Agreeableness:** cooperative, considerate, altruistic, sympathetic.
- **Conscientiousness:** organized, disciplined, hardworking, task-oriented.
- **Neuroticism:** emotionally unstable, prone to worry, anxiety, distress.
- **Extraversion:** sociable, outward-oriented, energetic, comfortable interacting with others.
- **Openness:** curious, imaginative, open to new ideas and experiences.

**Interpretation:** Personality measures are persistent over time and partly biologically rooted. This persistence reduces the concern that the paper is merely measuring contemporaneous portfolio shocks.

## 2.3 Beliefs

The survey elicits several belief measures:

- expected one-year stock market return;
- probability that the S&P 500 rises by more than 20%;
- probability that the S&P 500 falls by more than 20%;
- expected GDP growth;
- expected inflation.

**Interpretation:** These measures allow the authors to separate average return expectations from tail risk beliefs. This distinction is crucial because Neuroticism affects downside beliefs especially strongly.

## 2.4 Belief formation

The paper constructs an extrapolation score from two questions:

- if a stock has risen a lot over the last year, will it continue to rise, fall, remain the same, or is the respondent unsure?
- if a stock has fallen a lot over the last year, will it continue to fall, rise, remain the same, or is the respondent unsure?

A response consistent with continuation receives +100, a response consistent with reversal receives −100, and a neutral/unknown answer receives 0. The extrapolation score is the average of the two scores.

**Interpretation:** Higher scores mean more extrapolation and less mean reversion. This variable measures not the level of optimism, but how investors think trends evolve.

## 2.5 Risk aversion

Risk preferences are elicited using three job-choice questions. Each question asks the respondent to choose between a safe job and a risky job. The risky job always has a 50% chance of doubling income but has increasingly severe downside risk:

- Bet 1: income falls by 20% in the bad state;
- Bet 2: income falls by 33% in the bad state;
- Bet 3: income falls by 50% in the bad state.

The paper also constructs an implied risk-aversion parameter:

$$RiskAversion_i = \begin{cases} 1, & \text{takes all three risky jobs,} \\ 2, & \text{takes the first two but rejects the third,} \\ 3, & \text{takes only the first,} \\ 4, & \text{rejects all three.} \end{cases} \tag{1}$$

**Interpretation:** A higher value means more risk aversion. The elicited risk-aversion variable is largely uncorrelated with expected stock returns, so beliefs and risk preferences are distinct objects.

## 2.6 Social interaction tendency

The survey asks whether the respondent would consider investing in a new financial product if it becomes popular among people around them. The answer is coded from 1 to 5:

$$1 = \text{Definitely No}, \qquad 5 = \text{Definitely Yes}. \tag{2}$$

The paper also uses a binary version equal to one if the respondent says Yes or Definitely Yes.

**Interpretation:** This question captures a social herding tendency or willingness to adopt investments because they are popular in the respondent's social circle.

## 2.7 Portfolio choice

The main portfolio outcome is the equity share of financial wealth. The paper separately measures:

- total equity share;
- equity share in retirement accounts;
- equity share in non-retirement accounts.

**Interpretation:** Retirement and non-retirement accounts matter because portfolio composition can differ across account types and because non-retirement assets may include alternative risky assets not counted as equities.

## 2.8 External datasets: HILDA and GSOEP

- **HILDA:** the Household, Income and Labour Dynamics in Australia survey. Personality data are collected in 2005, 2009, 2013, and 2017; investment data are collected in 2002, 2006, 2010, and 2014. The paper matches adjacent waves.
- **GSOEP:** the German Socio-Economic Panel survey. Personality and investment data are collected in 2005, 2009, 2012, 2013, and 2017. The key outcome is stock market participation.

**Interpretation:** These datasets provide out-of-sample tests in more representative household panels and in different national contexts.

# 3 Conceptual framework

## 3.1 Investment decision problem

The model has a risk-free asset with zero interest rate and a stock with stochastic return $r$. Investor $i$ chooses an equity share $w_i$.

The investor maximizes a weighted objective:

$$\max_{w_i}(1-\alpha)\left(w_i E_i[r] - \frac{1}{2}\gamma_i w_i^2 Var_i(r)\right) - \alpha\frac{1}{2}(w_i - w_i^*)^2. \tag{1}$$

- $E_i[r]$ is the subjective expected stock return.
- $Var_i(r)$ is the subjective perceived risk.
- $\gamma_i$ is risk aversion.
- $w_i^*$ is the target portfolio from non-pecuniary motives.
- $\alpha \in [0,1]$ is the weight on non-pecuniary motives.

**Interpretation:** The model blends standard mean-variance utility with a target portfolio. The target portfolio captures social motives, excitement, ethical motives, inertia, or other nonstandard forces.

## 3.2 Optimal equity share

The first-order condition gives:

$$w_i = \frac{(1-\alpha)E_i[r] + \alpha w_i^*}{(1-\alpha)\gamma_i Var_i(r) + \alpha}. \tag{2}$$

**Interpretation:** Personality can affect portfolio choice through four channels: expected returns, perceived risk, risk aversion, and the target portfolio.

## 3.3 Special cases

If $\alpha = 0$, portfolio choice reduces to standard mean-variance demand:

$$w_i = \frac{E_i[r]}{\gamma_i Var_i(r)}. \tag{3}$$

If $\alpha = 1$, the portfolio is entirely determined by the target portfolio:

$$w_i = w_i^*. \tag{4}$$

**Interpretation:** The model naturally explains why survey-measured expected returns often have low sensitivity for portfolio shares: portfolio choice may partly reflect target-portfolio motives rather than pure return-risk optimization.

# 4 Empirical specifications

## 4.1 Personality and beliefs

For belief outcome $Y_i$, the generic specification is:

$$Y_i = \beta_1 Agree_i + \beta_2 Consc_i + \beta_3 Neuro_i + \beta_4 Extra_i + \beta_5 Open_i + \Gamma' X_i + \varepsilon_i. \tag{5}$$

$X_i$ includes demographic fixed effects: gender, age, income, wealth, education, and location.

**Interpretation:** This regression asks whether personality explains heterogeneity in beliefs after accounting for observable demographics.

## 4.2 Personality and risk aversion

For risk-taking variables and the implied risk-aversion parameter:

$$Risk_i = \beta' Personality_i + \Gamma' X_i + \varepsilon_i. \tag{6}$$

**Interpretation:** This regression asks whether personality traits map into conventional preference parameters.

## 4.3 Personality and social influence

For the social herding score or Yes/Definitely Yes dummy:

$$Social_i = \beta' Personality_i + \Gamma' X_i + \varepsilon_i. \tag{7}$$

**Interpretation:** This measures whether personality affects nonstandard investment motives outside beliefs and risk aversion.

## 4.4 Personality and equity allocation

The AAII portfolio regression is:

$$EquityShare_i = \beta' Personality_i + \Gamma' X_i + \varepsilon_i. \tag{8}$$

A richer version controls for beliefs and risk aversion:

$$EquityShare_i = \beta' Personality_i + \delta' Beliefs_i + \lambda RiskAversion_i + \Gamma' X_i + \varepsilon_i. \tag{9}$$

**Interpretation:** If personality coefficients remain significant after controlling for beliefs and risk aversion, this points to complementary survey measurement or nonstandard target-portfolio channels.

### 4.5 Panel specifications in HILDA and GSOEP

For the external panel datasets:

$$Y_{i,t} = \beta' Personality_{i,t} + \Gamma' X_{i,t} + YearFE_t + \varepsilon_{i,t}. \tag{10}$$

**Interpretation:** These regressions test whether the U.S. AAII findings survive in representative non-U.S. panels.

## 5 Identification and research design

### 5.1 What identifies the estimates

- The analysis is cross-sectional in the AAII survey and panel-based in HILDA/GSOEP.
- Identification relies on persistent individual personality variation after controlling for demographic differences.
- Personality traits are relatively stable over the life cycle, so they are less likely to be caused by contemporaneous portfolio choices.
- Out-of-sample validation in Australia and Germany helps address concerns about AAII sample specificity.

**Interpretation:** This is not a randomized design. The strength of the paper is measurement, mechanism, and replication across samples.

### 5.2 Why personality traits are useful beyond demographics

- Demographics explain only a small share of heterogeneity in beliefs and portfolios.
- Personality traits explain comparable variation in beliefs to the full set of demographic fixed effects.
- Neuroticism and Openness are domain-specific: they matter for finance even though other traits such as Agreeableness are important in labor-market applications.

**Interpretation:** Personality traits capture persistent psychological heterogeneity that standard household finance models often leave unmeasured.

### 5.3 Mechanism interpretation

- **Neuroticism:** pessimistic expected returns, higher crash probabilities, higher inflation expectations, mean-reverting beliefs, lower equity allocation.
- **Openness:** lower risk aversion, higher probabilities assigned to both tails, more extrapolative beliefs, higher equity allocation.
- **Extraversion:** more optimistic beliefs and stronger social-following tendency.
- **Agreeableness:** higher risk aversion but limited direct portfolio effect.
- **Conscientiousness:** more optimistic average expected returns and lower crash beliefs but weaker portfolio effect.

**Interpretation:** The traits do not all work the same way. Neuroticism primarily changes beliefs; Openness primarily changes risk tolerance and target-portfolio forces.

### 5.4 Limitations

- The AAII sample is affluent, older, mostly male, and more financially sophisticated than the general population.
- Survey responses may contain measurement error.
- Personality traits are persistent but not randomly assigned.
- The paper cannot fully separate all biological, social, and life-experience sources of personality.
- Portfolio outcomes are self-reported in the AAII survey.

**Interpretation:** The evidence is strongest as a disciplined measurement and mechanism study, not as an experiment.

# 6 Tables and figures

The tables and figures below are directly cropped from the paper and grouped compactly.

## 6.1 Table 1 and Table 2: sample summary and demographic correlates

**Read:**

- Table 1 reports that the AAII respondents are mostly male and white, with average age 68.23, average income $233,290, and average wealth $3.27 million.
- Personality scores vary meaningfully across respondents. Neuroticism has mean 3.39 and standard deviation 0.97; Openness has mean 4.48 and standard deviation 0.92.
- Average expected stock return is 5.57%, but the 10th percentile is −10% and the 90th percentile is 14%.
- Average predicted probability of a 20% market decline is 25.09%.
- Table 2 shows that demographics explain little of personality variation: adjusted $R^2$ values range from about 0.002 to 0.02.

**Economic message:** Personality traits are not reducible to standard demographic variables. They bring additional variation into household finance analysis.

**Table 1**
Summary statistics.

*Panel (a) Demographics and personality traits*

| | Mean | Std Dev | 10 Pct | 50 Pct | 90 Pct | Skewness |
|---|---|---|---|---|---|---|
| Male | 0.93 | 0.25 | 1.00 | 1.00 | 1.00 | -3.51 |
| White | 0.91 | 0.29 | 1.00 | 1.00 | 1.00 | -2.83 |
| Age | 68.23 | 8.50 | 55.00 | 75.00 | 75.00 | -1.43 |
| Income (in $1000) | 233.29 | 369.41 | 125.00 | 125.00 | 350.00 | 12.97 |
| Wealth (in $1000) | 3271.95 | 2353.79 | 750.00 | 3500.00 | 7500.00 | 0.76 |
| College | 0.90 | 0.30 | 0.00 | 1.00 | 1.00 | -2.65 |
| Agreeableness | 4.86 | 0.73 | 3.75 | 5.00 | 5.75 | -0.84 |
| Conscientiousness | 4.89 | 0.74 | 3.75 | 5.00 | 5.75 | -0.80 |
| Neuroticism | 3.39 | 0.97 | 2.00 | 3.50 | 4.75 | -0.06 |
| Extraversion | 2.59 | 1.04 | 1.25 | 2.50 | 4.00 | 0.39 |
| Openness | 4.48 | 0.92 | 3.25 | 4.50 | 5.65 | -0.63 |

*Panel (b) Correlation matrix*

| | Agreeableness | Conscientiousness | Neuroticism | Extraversion | Openness |
|---|---|---|---|---|---|
| Agreeableness | 1.00 | 0.21 | 0.01 | 0.14 | 0.18 |
| Conscientiousness | 0.21 | 1.00 | -0.07 | 0.12 | 0.24 |
| Neuroticism | 0.01 | -0.07 | 1.00 | -0.14 | -0.11 |
| Extraversion | 0.14 | 0.12 | -0.14 | 1.00 | 0.16 |
| Openness | 0.18 | 0.24 | -0.11 | 0.16 | 1.00 |

*Panel (c) Beliefs and preferences*

| | Mean | Std Dev | 10 Pct | 50 Pct | 90 Pct | Skewness |
|---|---|---|---|---|---|---|
| Expected Stock Return | 5.57 | 9.51 | -10.00 | 7.00 | 14.00 | -1.23 |
| Stock Rise by >20% | 18.49 | 16.25 | 1.00 | 15.00 | 40.00 | 1.54 |
| Stock Fall by >20% | 25.09 | 18.41 | 5.00 | 24.00 | 50.00 | 1.02 |
| GDP Growth | 1.97 | 1.31 | 1.00 | 2.00 | 3.00 | -0.88 |
| Inflation | 2.05 | 1.03 | 1.00 | 2.00 | 3.00 | 0.30 |
| Pick Risky Job 1 | 0.60 | 0.49 | 0.00 | 1.00 | 1.00 | -0.42 |
| Pick Risky Job 2 | 0.27 | 0.44 | 0.00 | 0.00 | 1.00 | 1.03 |
| Pick Risky Job 3 | 0.06 | 0.23 | 0.00 | 0.00 | 0.00 | 3.77 |

Panel (a) reports the summary statistics of personality traits and demographic variables. “Male” is the dummy variable which is 1 if the respondent is a male. “White” is the dummy variable which is 1 if the respondent’s self-identified race is white. “College” is the dummy variable which is 1 if the respondent has a bachelor’s degree or above. There are 3,325 respondents in total.

*Table 1: Summary statistics*

**Table 2**
Personality traits and investor characteristics.

| | (1) Agreeableness | (2) Conscientiousness | (3) Neuroticism | (4) Extraversion | (5) Openness |
|---|---|---|---|---|---|
| Female | 0.29*** | -0.02 | 0.25** | 0.06 | -0.04 |
| | (0.07) | (0.07) | (0.11) | (0.10) | (0.09) |
| Age | 0.01*** | -0.01*** | -0.01** | 0.01*** | -0.01** |
| | (0.002) | (0.002) | (0.002) | (0.002) | (0.002) |
| Log Income | 0.03 | 0.09*** | -0.08** | 0.09*** | 0.05 |
| | (0.02) | (0.02) | (0.03) | (0.03) | (0.03) |
| Log Wealth | -0.04* | 0.04** | -0.03 | 0.02 | 0.01 |
| | (0.02) | (0.02) | (0.03) | (0.03) | (0.02) |
| College | 0.05 | -0.03 | 0.03 | -0.08 | 0.07 |
| | (0.05) | (0.05) | (0.07) | (0.07) | (0.06) |
| Race F.E. | Y | Y | Y | Y | Y |
| State F.E. | Y | Y | Y | Y | Y |
| Observations | 2,607 | 2,607 | 2,607 | 2,607 | 2,607 |
| $R^2$ | 0.04 | 0.05 | 0.04 | 0.05 | 0.03 |
| Adjusted $R^2$ | 0.01 | 0.02 | 0.01 | 0.02 | 0.002 |

We regress each personality trait on demographic variables. In these regressions, we use the subsample of the AAII respondents who indicate they are either male or female, and provide their income and wealth information. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

*Table 2: Personality traits and investor characteristics*

## 6.2 Figures 1 and 2: distributions of demographics and personality traits

**Read:**

- Figure 1 shows that the AAII sample is skewed toward older, wealthier, more educated investors.
- Figure 2 shows meaningful heterogeneity in all five personality traits.
- Openness, Agreeableness, and Conscientiousness are negatively skewed, while Neuroticism is closer to symmetric and Extraversion is more right-skewed.

**Economic message:** The sample is selected, but there remains substantial psychological variation within the sample.

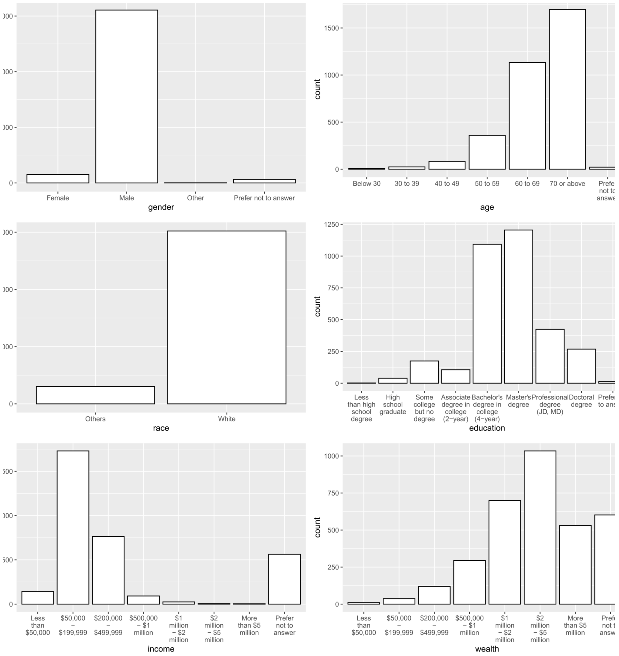



Fig. 1. Distribution of demographic variables in the AAII survey.

*Figure 1: Demographic distributions*

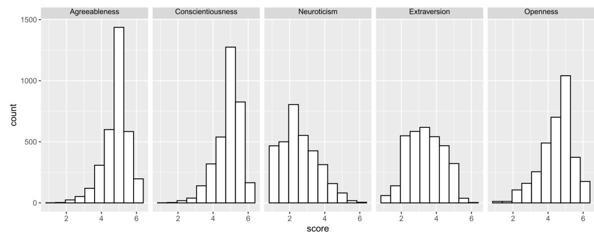



Fig. 2. Distribution of personality traits in the AAII survey.

*Figure 2: Personality trait distributions*

## 6.3 Table 3 and Table 4: beliefs and belief formation

**Read:**

- Table 3 shows that Neuroticism is associated with lower expected stock returns: a one-point increase predicts a 79-basis-point lower expected return.
- Neuroticism also predicts a 102-basis-point higher perceived probability of a 20% market crash.
- Extraversion predicts more optimistic expected returns and lower crash beliefs.
- Openness is not related to average expected returns, but it predicts higher probabilities for both upside and downside tails.
- Table 4 shows that Neuroticism predicts more mean-reverting beliefs, while Openness predicts more extrapolative beliefs.

**Economic message:** Personality traits shape not only optimism and pessimism but also the way investors interpret trends and tail risks.

**Table 3**
Personality traits and investor beliefs.

*Panel (a) Benchmark results*

| | (1) Stock Return Mean | (2) Prob(>20%) | (3) Prob(< −20%) | (4) GDP Growth Mean | (5) Inflation Mean |
|---|---|---|---|---|---|
| Agreeableness | -0.10 | -0.34 | -0.09 | -0.01 | 0.002 |
| | (0.24) | (0.40) | (0.46) | (0.03) | (0.03) |
| Conscientiousness | 0.66*** | -0.07 | -0.99** | 0.04 | -0.07*** |
| | (0.24) | (0.40) | (0.46) | (0.03) | (0.03) |
| Neuroticism | -0.79*** | -0.21 | 1.02*** | -0.07*** | 0.05*** |
| | (0.16) | (0.28) | (0.32) | (0.02) | (0.02) |
| Extraversion | 0.82*** | 1.27*** | -1.07*** | 0.09*** | -0.02 |
| | (0.18) | (0.30) | (0.34) | (0.02) | (0.02) |
| Openness | 0.04 | 1.49*** | 0.92** | -0.003 | 0.01 |
| | (0.19) | (0.32) | (0.37) | (0.03) | (0.02) |
| Demographics F.E. | Y | Y | Y | Y | Y |
| Observations | 3,325 | 3,325 | 3,325 | 3,325 | 3,325 |
| $R^2$ | 0.06 | 0.06 | 0.04 | 0.04 | 0.04 |
| Adjusted $R^2$ | 0.03 | 0.04 | 0.01 | 0.02 | 0.01 |

*Panel (b) Adjusted $R^2$ under alternative specifications of explanatory variables*

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| Personality Traits Only | 0.02 | 0.01 | 0.01 | 0.01 | 0.005 |
| Demographics F.E. Only | 0.01 | 0.02 | 0.01 | 0.01 | 0.01 |

Panel (a) reports the regressions of investor beliefs on personality traits. Each cell in Panel (b) reports the adjusted R-squared of a regression, with personality traits only or with demographics fixed effects only. Dependent variables are (1) the expected stock return, (2) the probability that the stock market rises by more than 20%, (3) the probability that the stock market falls by more than 20%, (4) the expected GDP growth rate, and (5) the expected inflation. Demographics fixed effects include gender, age, income, wealth, education and location. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

*Table 3: Personality traits and investor beliefs*

**Table 4**
Personality traits and belief formation.

| | (1) Extrapolation score |
|---|---|
| Agreeableness | 0.89 |
| | (0.86) |
| Conscientiousness | -0.38 |
| | (0.87) |
| Neuroticism | -1.30** |
| | (0.59) |
| Extraversion | -0.10 |
| | (0.64) |
| Openness | 1.55** |
| | (0.69) |
| Demographics F.E. | Y |
| Observations | 3,325 |
| $R^2$ | 0.03 |
| Adjusted $R^2$ | 0.01 |

This table reports results from an OLS regression, in which the dependent variable is a respondent's "extrapolation score" that is constructed based on her responses to the following two questions. 1) "If a stock's price has risen a lot over the last year, its price over the next year will..." 2) "If a stock's price has fallen a lot over the last year, its price over the next year will..." For the first question, a respondent receives a score of 100 if her answer is "Continue to rise;" a score of −100 if her answer is "Start to fall;" or a score of 0 if her answer is "Remain the same" or "Cannot say." Similarly, for the second question, a respondent receives a score of 100 if her answer is "Continue to fall;" a score of −100 if her answer is "Start to rise;" or a score of 0 if her answer is "Remain the same" or "Cannot say." A respondent's extrapolation score is the average of her scores for these two questions. Demographics fixed effects include gender, age, income, wealth, education and location. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

level of beliefs, but also the perception of trends and streaks. In general, the belief in mean-reversion or continuation in stock returns is not necessarily irrational. However, our evidence shows that the tendency of the belief in mean-reversion or continuation depends on personality traits, highlighting their important role in belief formation.

*4.2. Risk aversion*

*Table 4: Personality traits and belief formation*

## 6.4 Table 5 and Table 6: risk aversion and social influence

**Read:**

- Table 5 shows that Openness and Extraversion are associated with higher willingness to take risky jobs and lower implied risk aversion.
- Agreeableness is associated with lower willingness to take risky jobs and higher risk aversion.
- Neuroticism is weakly related to risk aversion but much more important for beliefs.
- Table 6 shows that Neuroticism and Extraversion are associated with stronger social-following tendencies.
- Neuroticism may capture fear of missing out, while Extraversion reflects enjoyment of social interaction.

**Economic message:** Personality affects both standard preference channels and nonstandard social channels of investment behavior.

**Table 5**
Personality traits and risk aversion.

| | (1) Bet 1 | (2) Bet 2 | (3) Bet 3 | (4) Risk aversion |
|---|---|---|---|---|
| Agreeableness | -0.03*** | -0.04*** | -0.01** | 0.09*** |
| | (0.01) | (0.01) | (0.01) | (0.02) |
| Conscientiousness | -0.01 | -0.01 | 0.002 | 0.02 |
| | (0.01) | (0.01) | (0.01) | (0.02) |
| Neuroticism | -0.01 | -0.02** | -0.002 | 0.03* |
| | (0.01) | (0.01) | (0.004) | (0.02) |
| Extraversion | 0.03*** | 0.03*** | 0.01 | -0.06*** |
| | (0.01) | (0.01) | (0.004) | (0.02) |
| Openness | 0.03*** | 0.03*** | 0.02*** | -0.08*** |
| | (0.01) | (0.01) | (0.005) | (0.02) |
| Demographics F.E. | Y | Y | Y | Y |
| Observations | 3,325 | 3,325 | 3,325 | 3,325 |
| $R^2$ | 0.06 | 0.05 | 0.03 | 0.06 |
| Adjusted $R^2$ | 0.04 | 0.02 | 0.003 | 0.04 |

In Columns (1)–(3), we regress the dummy variables indicating whether the respondent is willing to take each bet on personality traits and controls. In Column (4), the dependent variable is the implied risk aversion parameter from the survey responses. Demographics fixed effects include gender, age, income, wealth, education, and location. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

more conservative. Similarly, an investor with higher Extraversion enjoys social interaction and tends to be more excitement-seeking (McCrae and Costa Jr., 1997). However, the association between Agreeableness and risk aversion seems less obvious.

Conceptually, the results in Sections 4.1 and 4.2 suggest that personality traits can provide deeper psychological foundations for the origins of individual differences in beliefs and preferences (see McAdams, 2015, for a review). A related literature specifically examines how a particular genetic variation explains financial decisions through its effects on Neuroticism (Kuhnen et al., 2013, among others). Therefore, this could open up a new line of research that relates the origins of heterogeneous risk preference to personality traits, the biological and experiential foundations of which have been studied extensively in psychology and behavioral sciences.

*4.3. Social interaction tendencies*

A recent literature begins to investigate how social interactions contribute to financial decision-making (e.g., Bailey et al., 2018; Han et al., 2022; Hirshleifer, 2020). To capture this social aspect, we include the following question: "Upon seeing a new type of investment becoming popular among people around you, would you consider investing in it as well?" This captures a scenario that many investors face regularly—e.g., how to respond when Bitcoin became a popular investment amongst the general public—and the resulting measure can be interpreted as a measure of social "herding." The options range from "Definitely No" to

Table 5: Risk aversion

**Table 6**
Personality traits and social influence.

| | (1) Score | (2) Yes or definitely yes |
|---|---|---|
| Agreeableness | 0.01 | 0.001 |
| | (0.02) | (0.01) |
| Conscientiousness | 0.01 | -0.003 |
| | (0.02) | (0.01) |
| Neuroticism | 0.04*** | 0.01** |
| | (0.01) | (0.004) |
| Extraversion | 0.04*** | 0.01*** |
| | (0.01) | (0.004) |
| Openness | 0.02* | -0.002 |
| | (0.01) | (0.004) |
| Demographics F.E. | Y | Y |
| Observations | 3,325 | 3,325 |
| $R^2$ | 0.03 | 0.04 |
| Adjusted $R^2$ | 0.005 | 0.01 |

Column (1) reports the result from an OLS regression, in which the dependent variable is the score from 1 (Definitely No) to 5 (Definitely Yes) assigned by respondents to the question, "upon seeing a new type of investment becoming popular among people around you, would you consider investing in it as well?" In Column (2), we replace the dependent variable by the dummy variable indicating if the score is 4 (Yes) or 5 (Definitely Yes). Demographics fixed effects include gender, age, income, wealth, education and location. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

investment strategies (Han et al., 2022; Hirshleifer, 2020) and expectations (Bailey et al., 2018) transmit in the population, an aspect that has often been ignored in traditional finance models but has recently received growing attention.

**5. Personality traits and asset allocation**

Table 6: Social influence

## 6.5 Table 7: AAII equity allocation

**Read:**

- Table 7 is the main portfolio-choice result in the AAII data.
- High Neuroticism predicts lower total equity share and lower retirement-account equity share.
- Low Openness predicts lower equity share across total, retirement, and non-retirement accounts.
- Neuroticism remains significant after controlling for beliefs and risk aversion in total and retirement accounts.
- Expected return increases equity shares, downside tail beliefs reduce equity shares, and risk aversion reduces equity shares.

**Economic message:** Personality traits remain important even when traditional beliefs and risk preferences are included, suggesting omitted psychological channels or complementary measurement.

**Table 7**
Personality traits and equity allocation: AAII data.

| | (1) Total | (2) Retirement | (3) Non-Retirement | (4) Total | (5) Retirement | (6) Non-Retirement |
|---|---|---|---|---|---|---|
| Agreeableness | -0.46 | -0.02 | -0.70 | -0.39 | 0.12 | -0.61 |
| | (0.57) | (0.57) | (0.72) | (0.56) | (0.56) | (0.72) |
| Conscientiousness | -1.32** | -0.66 | -1.00 | -1.51*** | -0.84 | -1.17 |
| | (0.58) | (0.58) | (0.72) | (0.58) | (0.57) | (0.72) |
| Neuroticism | -1.74*** | -2.55*** | -0.80 | -1.44*** | -2.23*** | -0.55 |
| | (0.40) | (0.39) | (0.49) | (0.39) | (0.39) | (0.49) |
| Extraversion | -0.33 | 0.14 | -0.05 | -0.65 | -0.30 | -0.31 |
| | (0.43) | (0.43) | (0.53) | (0.43) | (0.42) | (0.54) |
| Openness | 0.94** | 1.50*** | 1.15** | 0.95** | 1.40*** | 1.14** |
| | (0.46) | (0.46) | (0.57) | (0.46) | (0.45) | (0.58) |
| Expected Return | | | | 0.23*** | 0.24*** | 0.22*** |
| | | | | (0.05) | (0.05) | (0.06) |
| Up Tail | | | | -0.01 | 0.04 | -0.02 |
| | | | | (0.03) | (0.03) | (0.04) |
| Down Tail | | | | -0.08*** | -0.09*** | -0.05 |
| | | | | (0.02) | (0.02) | (0.03) |
| Risk Aversion | | | | -1.17*** | -1.44*** | -0.90 |
| | | | | (0.44) | (0.44) | (0.56) |
| Demographic F.E. | Y | Y | Y | Y | Y | Y |
| Observations | 2,807 | 3,285 | 3,281 | 2,807 | 3,285 | 3,281 |
| $R^2$ | 0.08 | 0.07 | 0.09 | 0.10 | 0.10 | 0.10 |
| Adjusted $R^2$ | 0.05 | 0.05 | 0.07 | 0.07 | 0.07 | 0.08 |

Regression results based on our AAII survey. We regress each investor's equity-to-wealth ratio on personality traits and controls. Demographics fixed effects include gender, age, income, wealth, education, and location. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

## 6.6 Table 8 and Table 9: international out-of-sample evidence

**Read:**

- Table 8 uses the Australian HILDA panel and finds that Neuroticism negatively predicts equity share, while Openness positively predicts equity share.
- The HILDA evidence holds both in one-person households and among respondents responsible for household financial decisions.
- Table 9 uses the German GSOEP panel and studies stock market participation.
- Neuroticism negatively predicts stock market participation and Openness positively predicts participation.
- The results are qualitatively consistent across the U.S., Australia, and Germany.

**Economic message:** The key traits, Neuroticism and Openness, are not sample-specific. They replicate in representative household panels in other countries.

**Table 8**
Personality traits and equity allocation: Australian HILDA data.

| | One-person household (1) | Decision maker in the household (2) |
|---|---|---|
| Agreeableness | 0.04 | -0.17 |
| | (0.30) | (0.23) |
| Conscientiousness | -0.39 | -0.35* |
| | (0.27) | (0.20) |
| Neuroticism | -0.56** | -0.46** |
| | (0.27) | (0.20) |
| Extraversion | 0.13 | -0.26 |
| | (0.24) | (0.18) |
| Openness | 0.81*** | 0.63*** |
| | (0.25) | (0.20) |
| Demographic F.E. | Y | Y |
| Year F.E. | Y | Y |
| Observations | 5,542 | 8,924 |
| $R^2$ | 0.17 | 0.16 |
| Adjusted $R^2$ | 0.17 | 0.16 |

Regression results based on the HILDA survey, which has a panel structure. The dependent variable is the share of stock assets in households' total financial wealth, which is between 0 and 100. In Column (1), we use the subsample of one-person households. In Column (2), we use the subsample of respondents who claim to "always" or "usually" be the one who makes the household's savings, investment and borrowing decisions. Demographics fixed effects include gender, age, income, wealth, and location. We also control for year fixed effects. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

*Table 8: Australian HILDA data*

**Table 9**
Personality traits and equity allocation: German GSOEP data.

| | One-person household (1) | Decision maker in the household (2) |
|---|---|---|
| Agreeableness | 0.30 | -0.73 |
| | (0.40) | (0.45) |
| Conscientiousness | -2.06*** | -1.97*** |
| | (0.61) | (0.56) |
| Neuroticism | -1.07** | -0.94*** |
| | (0.38) | (0.28) |
| Extraversion | -1.16** | -1.11* |
| | (0.41) | (0.54) |
| Openness | 1.11*** | 1.27*** |
| | (0.25) | (0.35) |
| Demographic F.E. | Y | Y |
| Year F.E. | Y | Y |
| Observations | 10,250 | 10,781 |
| $R^2$ | 0.15 | 0.16 |
| Adjusted $R^2$ | 0.15 | 0.15 |

Regression results based on the GSOEP survey, which has a panel structure. The dependent variable is stock market participation, which is 0 if the person holds no stock assets and 100 if the person holds any stock assets. In Column (1), we use the subsample of one-person households. In Column (2), we use the subsample of respondents who claim to be the head of household. Demographics fixed effects include gender, age, income, wealth, and location. We also control for year fixed effects. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

*Table 9: German GSOEP data*

## 6.7 Figure 3: personality and the seven investor characteristics

**Read:**

- Figure 3 groups respondents by Neuroticism or Openness and plots seven characteristics: expected returns, risk aversion, perceived up-tail risk, perceived down-tail risk, extrapolation score, social interaction tendency, and equity allocation.
- The figure shows that Neuroticism and Openness organize several dimensions of investor heterogeneity at once.
- The paper also runs a PCA of the seven characteristics. PC1 explains 22% of the total variance and resembles optimism. PC2 explains 18% and resembles openness to tails, low risk aversion, and social interaction.

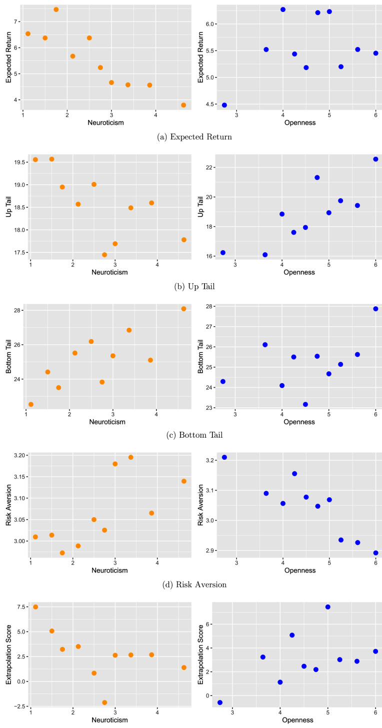
**Economic message:** The same two personality traits summarize common components in beliefs, preferences, social interaction, and portfolio choice.

## 6.8 Appendix Table A1: explanatory power for beliefs

**Read:**

- Table A1 compares the adjusted $R^2$ from personality traits and demographics in explaining investor beliefs.
- The explanatory power of the five personality traits is similar to that of demographic variables.
- Neuroticism and Openness alone have explanatory power close to the full set of five personality traits for several belief outcomes.

**Economic message:** The appendix reinforces the main argument: personality traits explain belief heterogeneity that demographics alone do not absorb.



(a) Expected Return

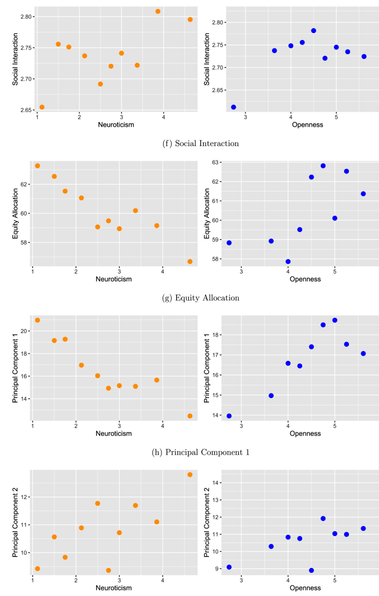
(b) Up Tail

(c) Bottom Tail

(d) Risk Aversion

(e) Extrapolation Score

**Fig. 3.** Investor characteristics vs. neuroticism and openness.



(f) Social Interaction

(g) Equity Allocation

(h) Principal Component 1

(i) Principal Component 2

**Fig. 3.** (*continued*)

**Table A1**
Explanatory power of different variables for investor belief.

| | (1) Market 30 Day | (2) Market 1 Year | (3) Self 30 Day | (4) Self 1 Year |
|---|---|---|---|---|
| Demographics F.E. Only | 0.008 | 0.027 | 0.029 | 0.042 |
| Personality Traits Only | 0.015 | 0.027 | 0.020 | 0.022 |
| Neuroticism and Openness Only | 0.012 | 0.019 | 0.020 | 0.019 |

We regress investor beliefs on either demographic variables or personality traits. Each cell reports the adjusted R-squared of a regression. The dependent variable is the expected market return in the next 30 days or the next year, or the expected return of the investor's own portfolio in the next 30 days or the next year, in columns (1) through (4), respectively. The independent variables are demographics fixed effects (including gender, age, income, wealth, and education) in the first row, the Big Five personality traits in the second row, and traits Neuroticism and Openness in the third row.

sehold wealth (including real estate, finan-
ns, etc.)?

**mpirical results on investor belief**

scribe the additional survey we ran among e administered the survey through the In- the Shenzhen Stock Exchange (SZSE). The in Jiang et al. (2022), which includes more tshell, we randomized across branch offices s. Specifically, we selected 2,993 branch of- nd regions) and required each branch office esponses.

etween November 29, 2021, and January 6, e given two weeks. A valid response had to inutes. Respondents could open the survey ters or on their smartphones; the vast major- ies. After applying basic filters, we collected 17,324 respondents. By design, respondents s the 60 brokers, with only slight variation. iation, areas that are more financially de- Zhejiang, Jiangsu, and Shanghai) are more mple is young, well-educated, and affluent: 5, the majority have a bachelor degree, and a wealth above 1 million RMB.

mented the same 20-item personality ques- into Chinese. We also asked the respondents the stock market's performance in the next r, as well as their expectations of their own ce in the next 30 days and in the next year. l variables, including age, gender, level of d total income, which we refer to below as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# 7 Short summary

- The paper introduces personality traits as persistent psychological characteristics relevant for household finance.
- The authors survey 3,325 AAII investors and measure the Big Five traits, beliefs, risk preferences, social tendencies, and portfolio shares.
- Neuroticism is associated with pessimistic expected stock returns, higher crash probabilities, lower GDP growth expectations, and higher inflation expectations.
- Openness is associated with lower risk aversion and higher probabilities assigned to both up and down tail events.
- Neuroticism predicts more mean-reverting beliefs; Openness predicts more extrapolative beliefs.
- Neuroticism and Extraversion predict stronger social-following tendencies.
- High Neuroticism and low Openness predict lower equity allocations.
- These portfolio effects remain after controlling for measured beliefs and risk aversion.
- The results replicate in HILDA and GSOEP, with Neuroticism negatively and Openness positively related to equity investment or stock market participation.
- The broad lesson is that personality traits provide a unified way to understand heterogeneity in beliefs, preferences, social interaction, and investment decisions.

# 8 Conclusion

## 8.1 Core conclusion

Jiang, Peng, and Yan show that personality traits help explain how investors form beliefs, perceive risk, respond to social environments, and allocate wealth to equities. Neuroticism and Openness are the key traits. Neurotic investors invest less in equities partly because they are more pessimistic and more worried about crashes. Open investors invest more in equities partly because they are less risk-averse and more willing to entertain new or extreme possibilities.

## 8.2 Economic intuition

The model separates portfolio choice into a standard mean-variance channel and a nonstandard target-portfolio channel. Personality can enter both. Neuroticism mostly shifts beliefs toward pessimism and downside worries. Openness mostly shifts risk tolerance and target-portfolio motives. This explains why personality remains significant even after controlling for standard belief and risk-preference measures.

## 8.3 Takeaway

The paper expands household finance beyond demographics, wealth, and financial literacy. Personality traits are stable, measurable, and economically meaningful. They help explain why investors with similar resources and information can hold very different beliefs and portfolios.